

Name – Ganesh Malaji Ghule
Roll No - 37

Assignment No 2

- Data Wrangling II

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```
In [1]: import pandas as pd
import numpy as np

# Generate sample data
np.random.seed(0)
data = {
    'Student_ID': range(1, 101),
    'Math': np.random.randint(50, 100, 100).astype(object),
    'Physics': np.random.randint(40, 95, 100),
    'English': np.random.randint(30, 90, 100).astype(object),
    'Attendance': np.random.uniform(0.7, 1, 100).astype(object),
    'GPA': np.random.uniform(2.5, 4, 100)
}

# Introduce some missing values and inconsistencies
data['Math'][10] = np.nan
data['Physics'][20] = 200 # introducing an outlier
data['English'][30] = 'A' # introducing inconsistency

# Create DataFrame
df = pd.DataFrame(data)
```

In [2]: df

Out[2]:

	Student_ID	Math	Physics	English	Attendance	GPA
0	1	94	45	63	0.983236	3.573342
1	2	97	81	70	0.882476	3.229239
2	3	50	75	62	0.878997	3.562822
3	4	53	40	66	0.935093	3.247210
4	5	53	71	89	0.850008	3.766825
...	...	...	...	...	...	...

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```
97 98 99 100 0.836551 3.597096
98 99 98 93 35 0.963661 3.650445
99 100 99 59 83 0.84768 2.514650

100 rows x 6 columns
```

```
In [3]: missing_values = df.isnull().sum()
print("Missing Values:\n", missing_values)

Missing Values:
Student_ID    0
Math          1
Physics       0
English       0
Attendance    0
GPA           0
dtype: int64
```

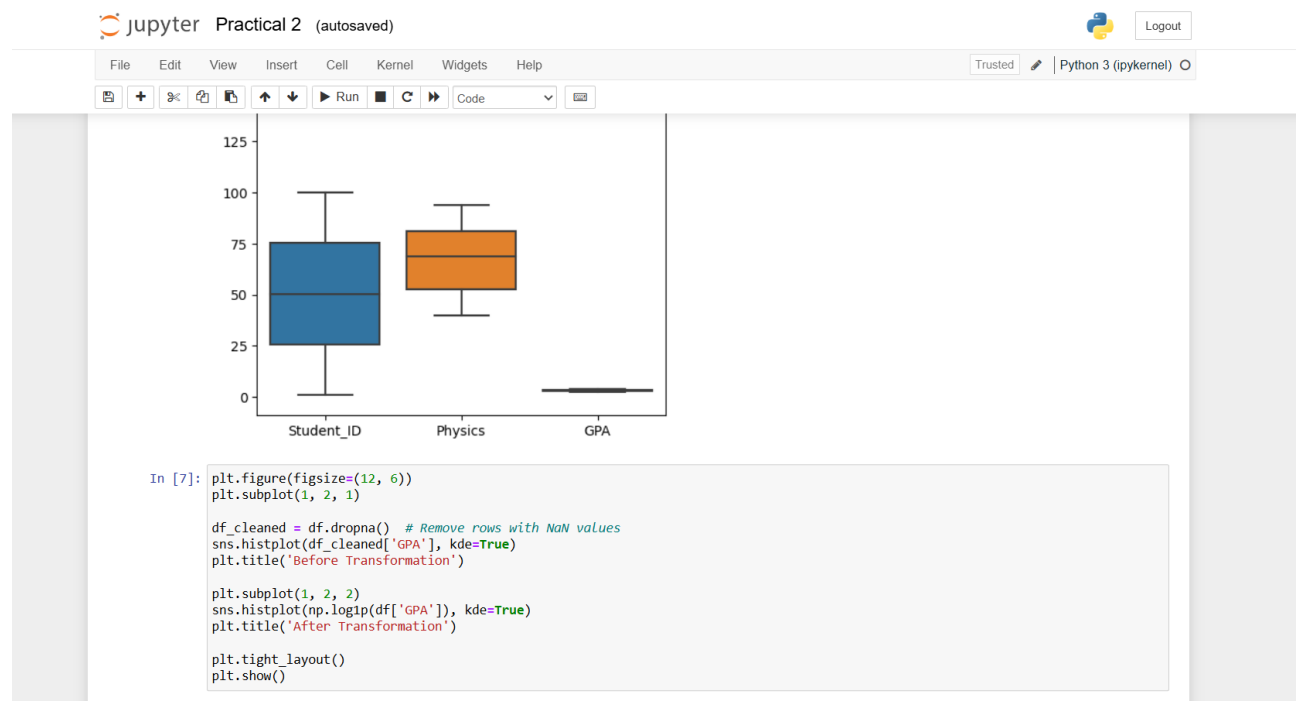
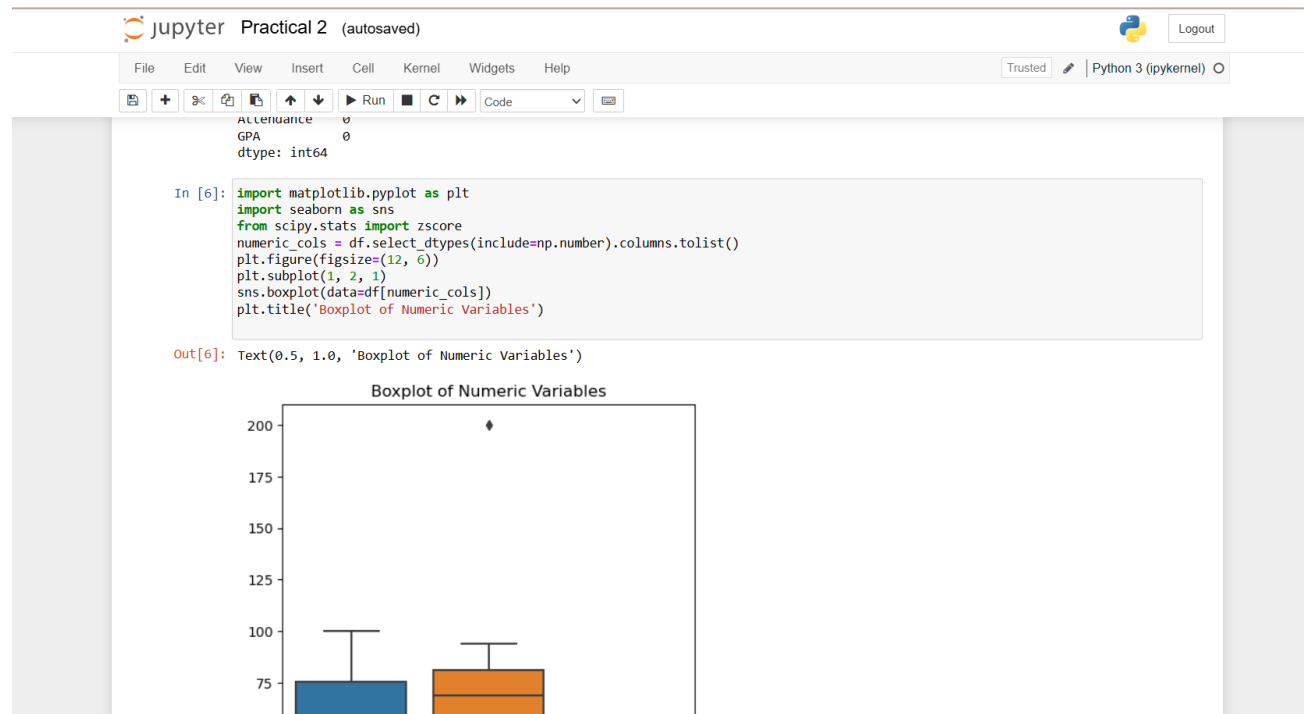
```
In [4]: # Check for inconsistencies in 'English_Score'
inconsistent_data = pd.to_numeric(df['English'], errors='coerce').isnull().sum()
print("Inconsistencies in English_Score:", inconsistent_data)

Inconsistencies in English_Score: 1
```

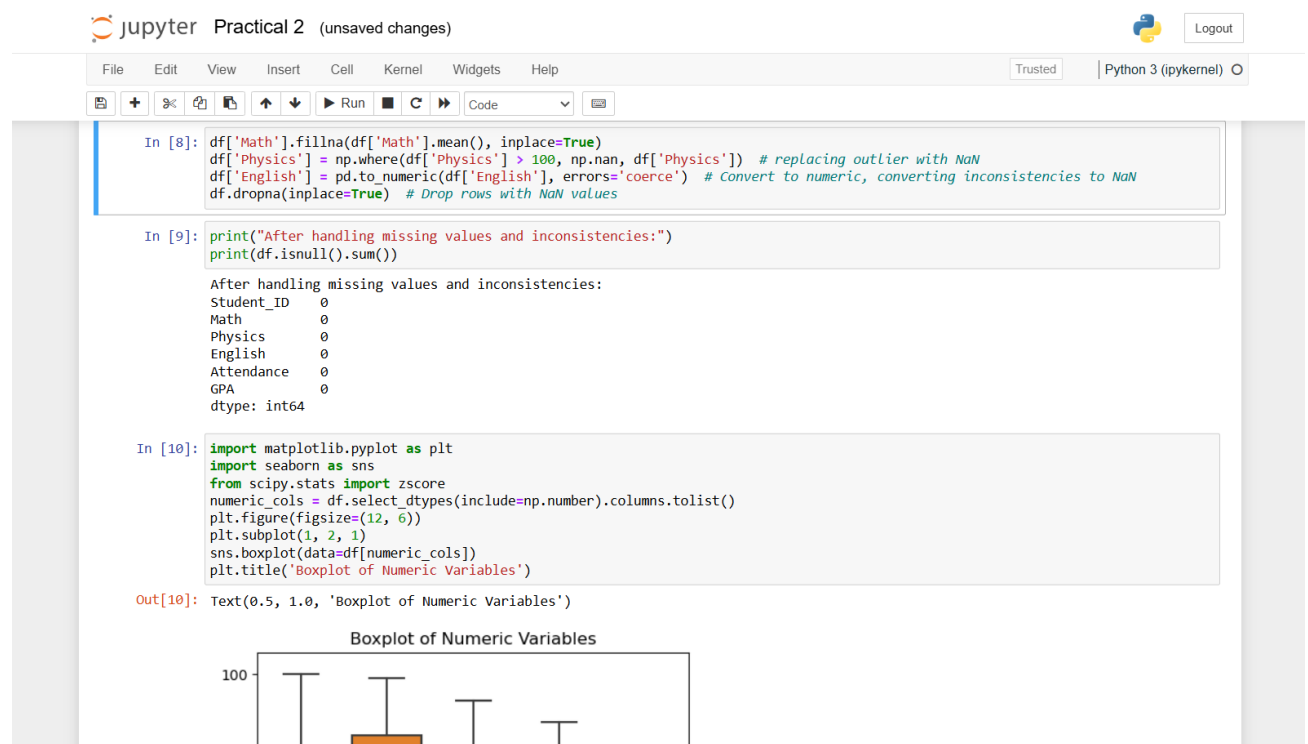
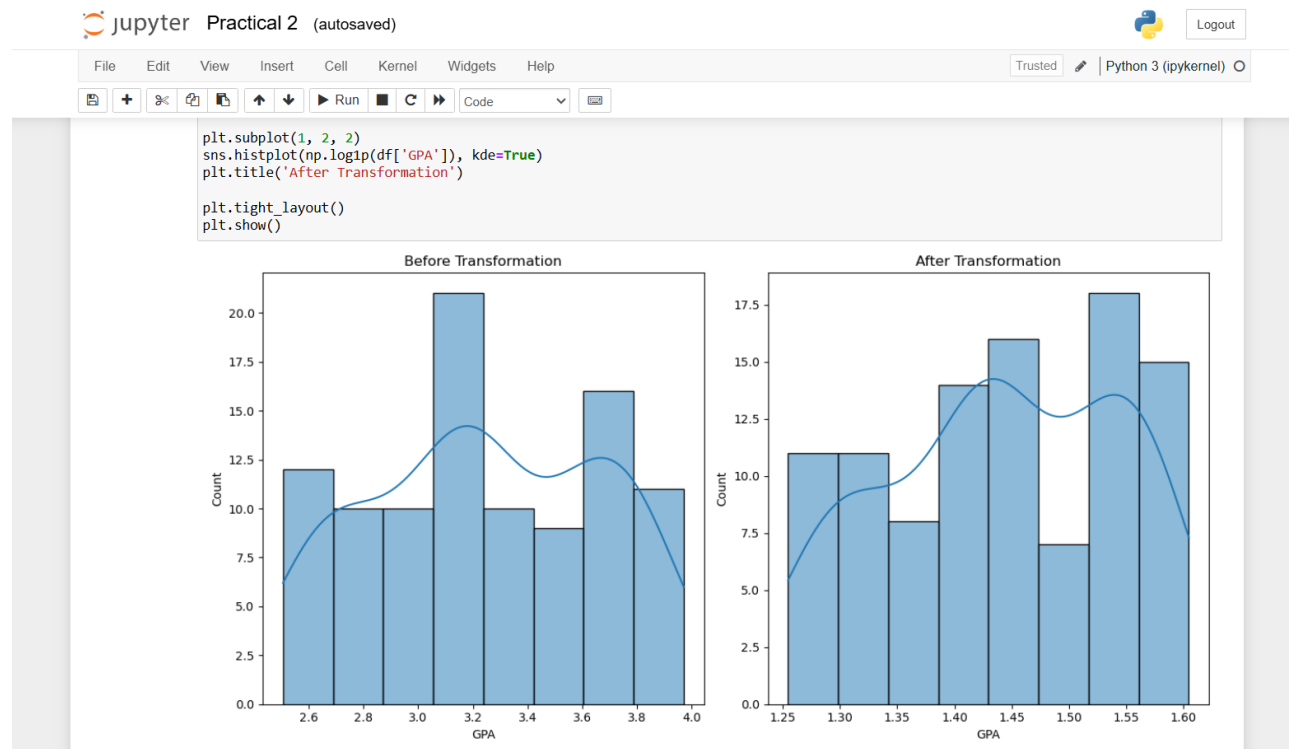
```
In [5]: print(df.isnull().sum())

Student_ID    0
Math          1
Physics       0
English       0
Attendance    0
GPA           0
dtype: int64
```

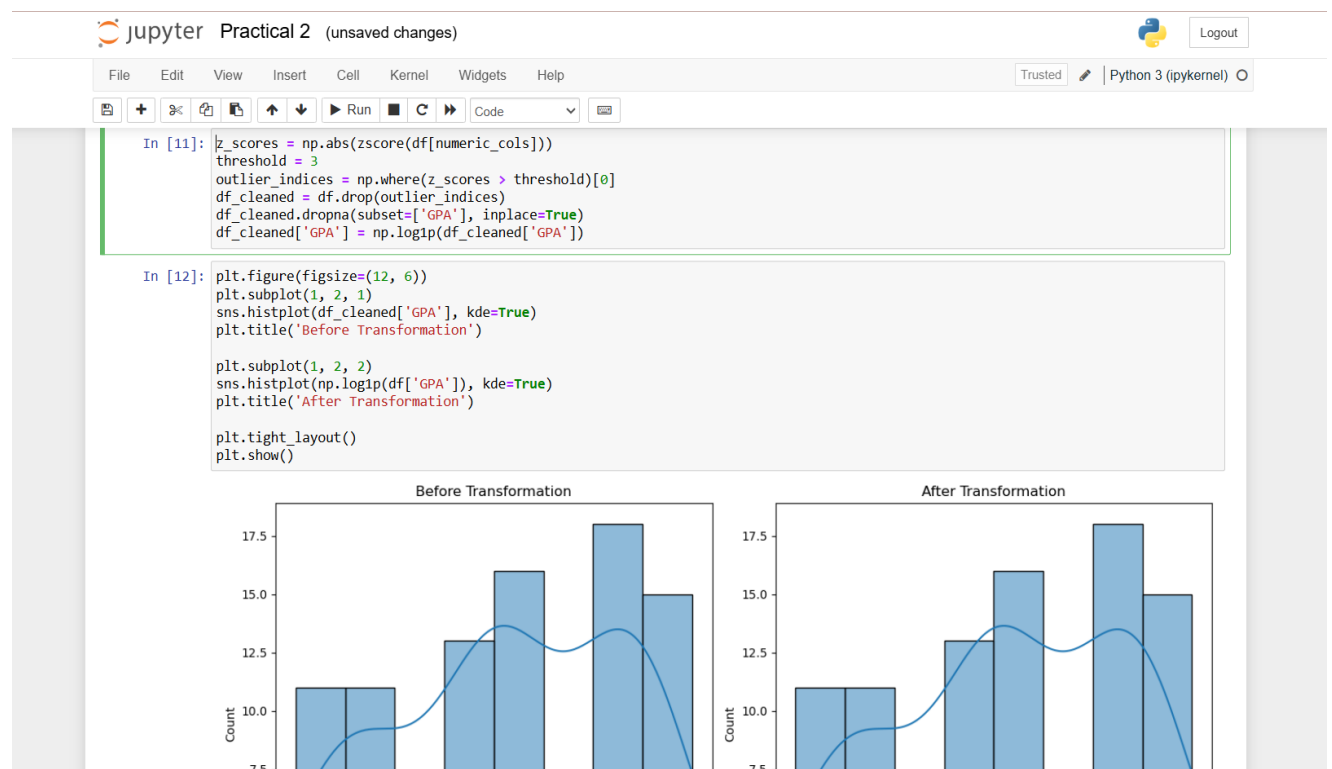
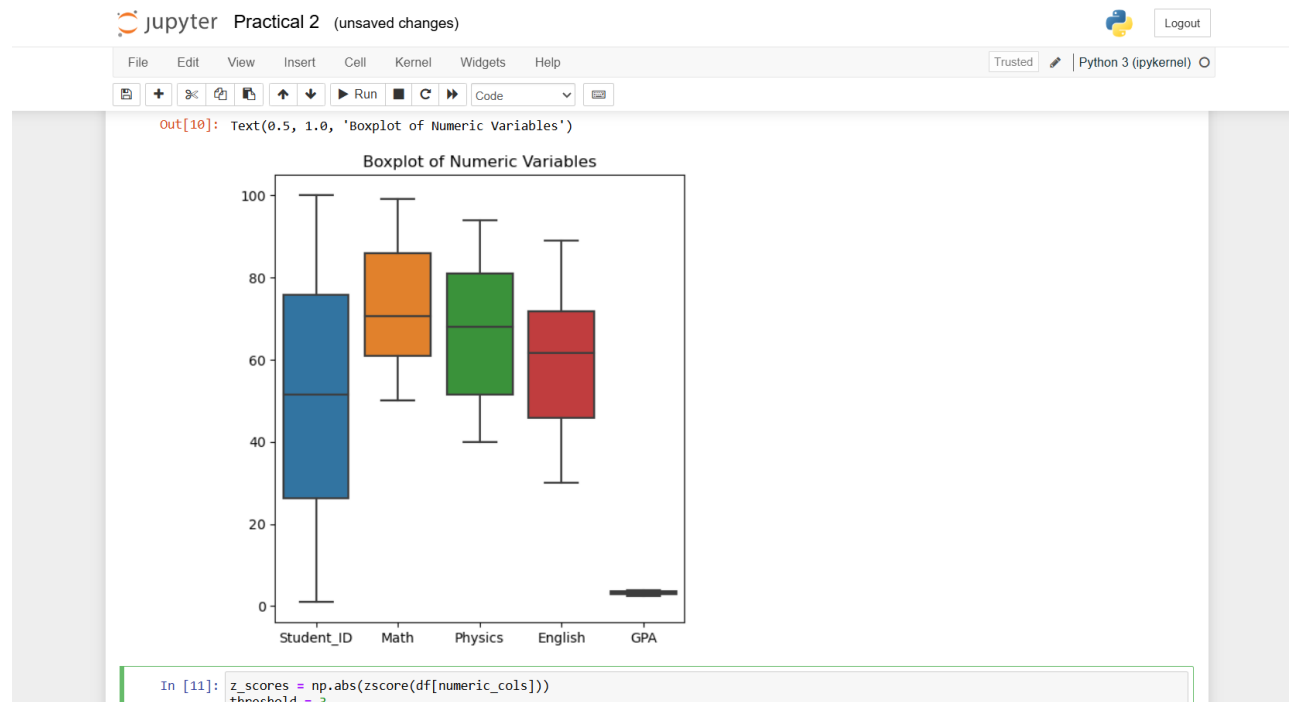
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